

Laporan Praktikum Jaringan Komputer WEEK 3

HTTP

NAMA : ANANDHAKA GHIBRAN MAS

NIM : 103072400135

Tujuan Praktikum :

1. Mahasiswa dapat menginvestigasi cara kerja protokol HTTP menggunakan Wireshark.

Basic HTTP GET/response interaction

gaia.cs.umass.edu/wireshark-labs/ HTTP-wireshark-file1.html

Congratulations. You've downloaded the file <http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html>!

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

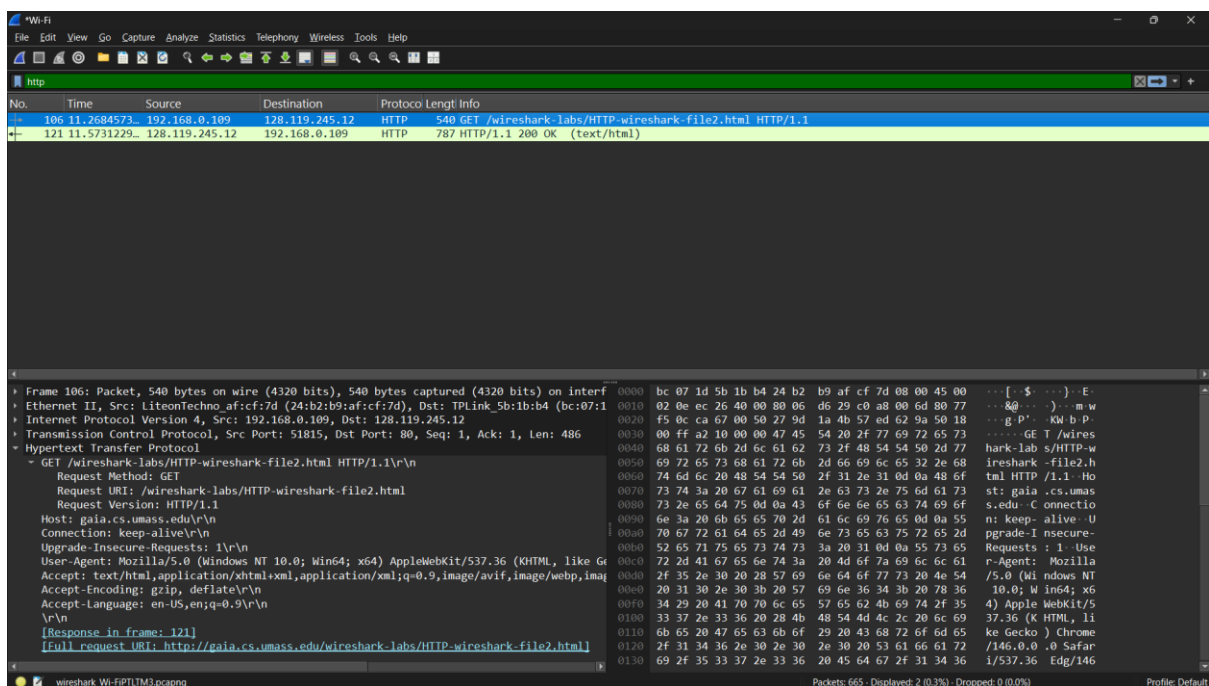
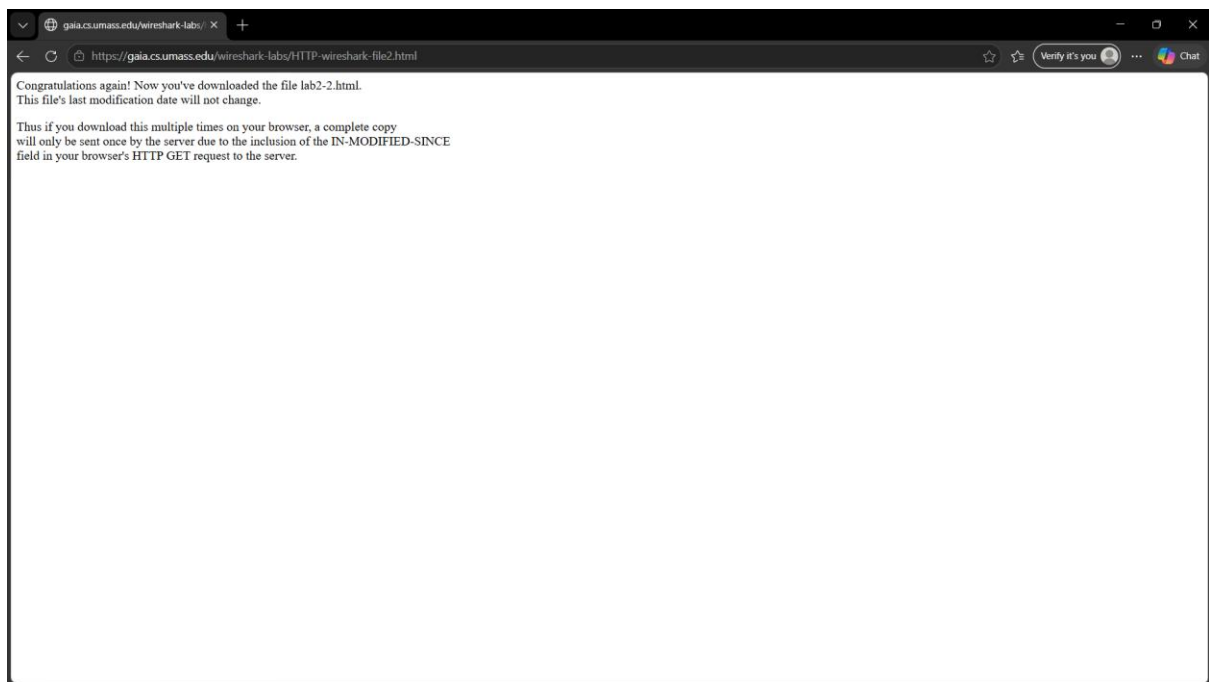
No.	Time	Source	Destination	Protocol	Length	Info
514	4.835969800	192.168.0.109	128.119.245.12	HTTP	548	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
522	5.243785700	128.119.245.12	192.168.0.109	HTTP	495	HTTP/1.1 200 OK (text/html)
529	5.272738500	192.168.0.109	128.119.245.12	HTTP	494	GET /favicon.ico HTTP/1.1
536	5.743470100	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)

Frame 514: Packet, 548 bytes on wire (4384 bits), 548 bytes captured (4384 bits) on interface 0
Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink Sb:1b:b4 (bc:07:11:1b:b4)
Internet Protocol Version 4, Src: 192.168.0.109, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 64037, Dst Port: 80, Seq: 1, Ack: 1, Len: 494
Hypertext Transfer Protocol
GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
Host: gaia.cs.umass.edu
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/106.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/svg+xml,*/*;q=0.8
Accept-Encoding: gzip, deflate
Accept-Language: id-ID;q=0.9,en-US;q=0.8,en;q=0.7
[Response in frame 522]
[Full request URL: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]

HTTP User-Agent header (http.user_agent), 125 bytes

Packets: 1210 - Displayed: 4 (0.3%) - Dropped: 0 (0.0%) Profile: Default

HTTP CONDITIONAL GET/response interaction



Retrieving Long Documents

gaia.cs.umass.edu/wireshark-labs/ Historical Documents: THE BILL OF Rights

https://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file3.html

THE BILL OF RIGHTS
Amendments 1-10 of the Constitution

The Conventions of a number of the States having, at the time of adopting the Constitution, expressed a desire, in order to prevent misconstruction or abuse of its powers, that further declaratory and restrictive clauses should be added, and as extending the ground of public confidence in the Government will best insure the beneficent ends of its institution;

Resolved, by the Senate and House of Representatives of the United States of America, in Congress assembled, two-thirds of both Houses concurring, that the following articles be proposed to the Legislatures of the several States, as amendments to the Constitution of the United States; all or any of which articles, when ratified by three-fourths of the said Legislatures, to be valid to all intents and purposes as part of the said Constitution, namely:

Amendment I

Congress shall make no law respecting an establishment of religion, or prohibiting the free exercise thereof; or abridging the freedom of speech, or of the press; or the right of the people peaceably to assemble, and to petition the government for a redress of grievances.

Amendment II

A well regulated militia, being necessary to the security of a free state, the right of the people to keep and bear arms, shall not be infringed.

Amendment III

No soldier shall, in time of peace be quartered in any house, without the consent of the owner, nor in time of war, but in a manner to be prescribed by law.

Amendment IV

The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no warrants shall issue, but upon probable cause, supported by oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized.

Amendment V

No person shall be held to answer for a capital, or otherwise infamous crime, unless on a presentment or indictment of a grand jury, except in cases arising in the land or naval forces, or in the militia, when in actual service in time of war or public danger; nor shall any person be subject for the same offense to be twice put in jeopardy of life or limb; nor shall be compelled in any criminal case to be a witness against himself, nor be deprived of life, liberty, or property, without due process of law; nor shall private property be taken for public use, without just compensation.

Amendment VI

In all criminal prosecutions, the accused shall enjoy the right to a speedy and public trial, by an impartial jury of the state and district wherein the crime shall have been committed, which district shall have been previously ascertained by law, and to be informed of the nature and cause of the accusation; to be confronted with the witnesses against him; to have compulsory process for obtaining witnesses in his favor, and to have the assistance of counsel for his defense.

Amendment VII

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
63	7.675163800	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
112	8.017313100	128.119.245.12	192.168.0.109	HTTP	562	HTTP/1.1 200 OK (text/html)

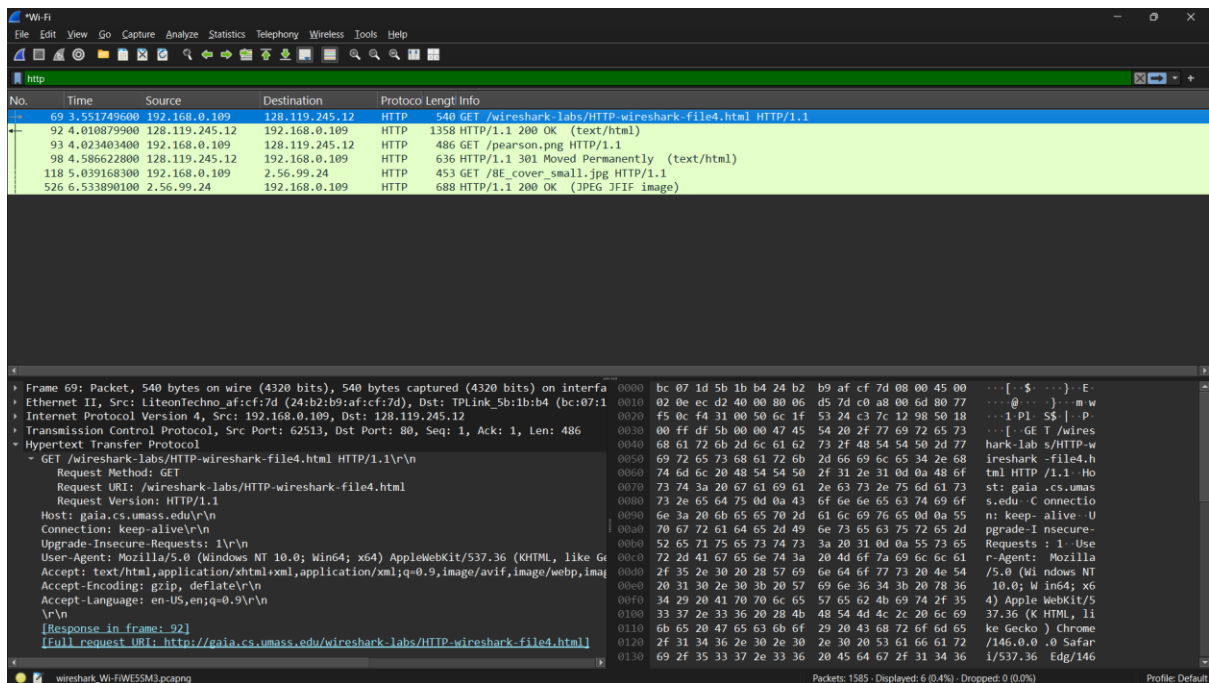
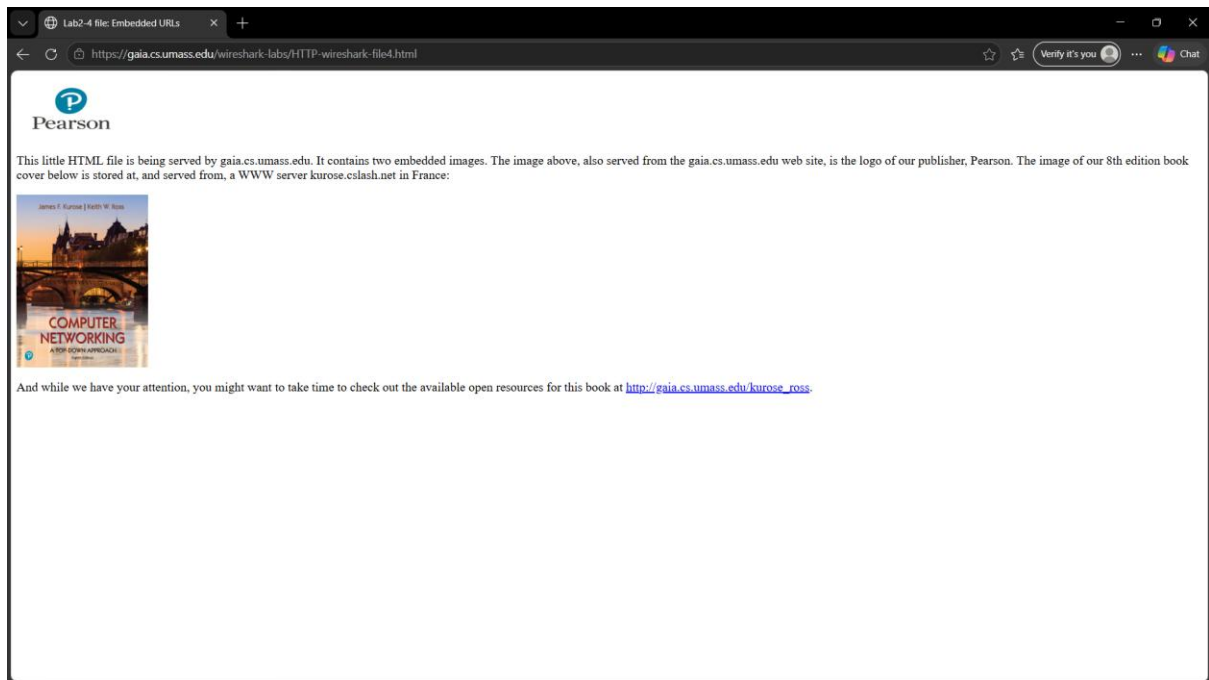
Frame 63: Packet, 540 bytes on wire (4320 bits), 540 bytes captured (4320 bits) on interface 0
Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_Sb:1b:b4 (bc:07:11:1b:b4:00:00)
Internet Protocol Version 4, Src: 192.168.0.109, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 49596, Dst Port: 80, Seq: 1, Ack: 1, Len: 486
Hypertext Transfer Protocol
GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
Request Method: GET
Request URI: /wireshark-labs/HTTP-wireshark-file3.html
Request Version: HTTP/1.1
Host: gaia.cs.umass.edu
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/146.0.0 Edg/146

Wi-Fi: <live capture in progress>

Packets: 439 - Displayed: 2 (0.5%)

Profile: Default

HTML Documents dengan Embedded Objects



Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
69	3.551749600	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
92	4.010879900	128.119.245.12	192.168.0.109	HTTP	1358	HTTP/1.1 200 OK (text/html)
93	4.023403400	192.168.0.109	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
98	4.586622800	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
118	5.039168300	192.168.0.109	2.56.99.24	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
526	6.533890100	2.56.99.24	192.168.0.109	HTTP	688	HTTP/1.1 200 OK (JPEG JFIF image)

Frame 93: Packet, 486 bytes on wire (3888 bits), 486 bytes captured (3888 bits) on interface 0000 bc 07 1d 5b 1b b4 24 b2 b9 af cf 7d 08 00 45 00 ...[...\$...]-E-

Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink 5b:1b:b4 (bc:07:1 0010 01 d8 ec d3 40 00 80 06 d5 b2 c0 a8 00 6d 80 77 ...@...m w

Internet Protocol Version 4, Src: 192.168.0.109, Dst: 128.119.245.12 0020 f5 0c f4 31 00 50 6c 1f 55 0a c3 7c 17 b0 50 18 ...1.P1: U...P-

Transmission Control Protocol, Src Port: 62513, Dst Port: 80, Seq: 487, Ack: 1305, Len: 43 0030 00 fa e2 ce 00 00 47 45 54 20 2f 70 65 61 72 73 ...-GE T /pears

Hypertext Transfer Protocol 0040 6f 6e 2e 70 6e 67 20 48 54 54 50 2f 31 2e 31 0d on.png H TTP/1.1-

Request Method: GET 0050 0a 4b 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 -Host: g aia.cs.u

Request URI: /pearson.png 0060 6d 61 73 73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 mass.edu -Connec

Request Version: HTTP/1.1 0070 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 tion: ke ep-alive

Host: gaia.cs.umass.edu\r\n 0080 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f -User-A gent: Mo

Connection: keep-alive\r\n 0090 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f zilla/5. 0 (Windo

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like G 00a0 77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 ws NT 10.0; Win6

Accept: image/avif,image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8\r\n 00b0 34 3b 20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4; x64) AppleWeb

Referer: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html\r\n 00c0 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d Kit/537. 36 (KHTM

Accept-Encoding: gzip, deflate\r\n 00d0 4c 2c 20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 L, like Gecko) C

Accept-Language: en-US,en;q=0.9\r\n 00e0 68 72 6f 6d 65 2f 31 34 36 2c 30 2e 30 2e 30 20 hrome/14 6.0.0.0

\r\n 00f0 53 61 66 61 72 69 2f 35 33 37 2e 33 36 20 45 64 Safari/5 37.36 Ed

[Response in frame: 98] 0100 67 2f 31 34 36 2e 30 2e 30 2e 30 0d 0a 41 63 63 g/146.0. 0.0 Acc

[Full request URI: http://gaia.cs.umass.edu/pearson.png] 0110 65 70 74 3a 20 69 6d 61 67 65 2f 61 76 69 66 2c ept: ima ge/avif,

0120 69 6d 61 67 65 2f 77 65 62 70 2c 69 6d 61 67 65 image/we bp,image

0130 2f 61 70 6e 67 2c 69 6d 61 67 65 2f 73 76 67 2b /apng,im age/svg+

Packets: 1585 - Displayed: 6 (0.4%) - Dropped: 0 (0.0%) Profile: Default

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
69	3.551749600	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
92	4.010879900	128.119.245.12	192.168.0.109	HTTP	1358	HTTP/1.1 200 OK (text/html)
93	4.023403400	192.168.0.109	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
98	4.586622800	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
118	5.039168300	192.168.0.109	2.56.99.24	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
526	6.533890100	2.56.99.24	192.168.0.109	HTTP	688	HTTP/1.1 200 OK (JPEG JFIF image)

Frame 93: Packet, 486 bytes on wire (3888 bits), 486 bytes captured (3888 bits) on interface 0
Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:11:5b:1b:b4)
Internet Protocol Version 4, Src: 192.168.0.109, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 62513, Dst Port: 80, Seq: 487, Ack: 1305, Len: 43
Hypertext Transfer Protocol
GET /pearson.png HTTP/1.1
Request Method: GET
Request URI: /pearson.png
Request Version: HTTP/1.1
Host: gaia.cs.umass.edu
Connection: keep-alive
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/69.0.3497.100 Safari/537.36
Accept: image/avif,image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8
Referer: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9
[Response in frame: 98]
[Full request URI: http://gaia.cs.umass.edu/pearson.png]

Wireshark Wi-FiWESSM3.pcapng

Packets: 1585 - Displayed: 6 (0.4%) - Dropped: 0 (0.0%)

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

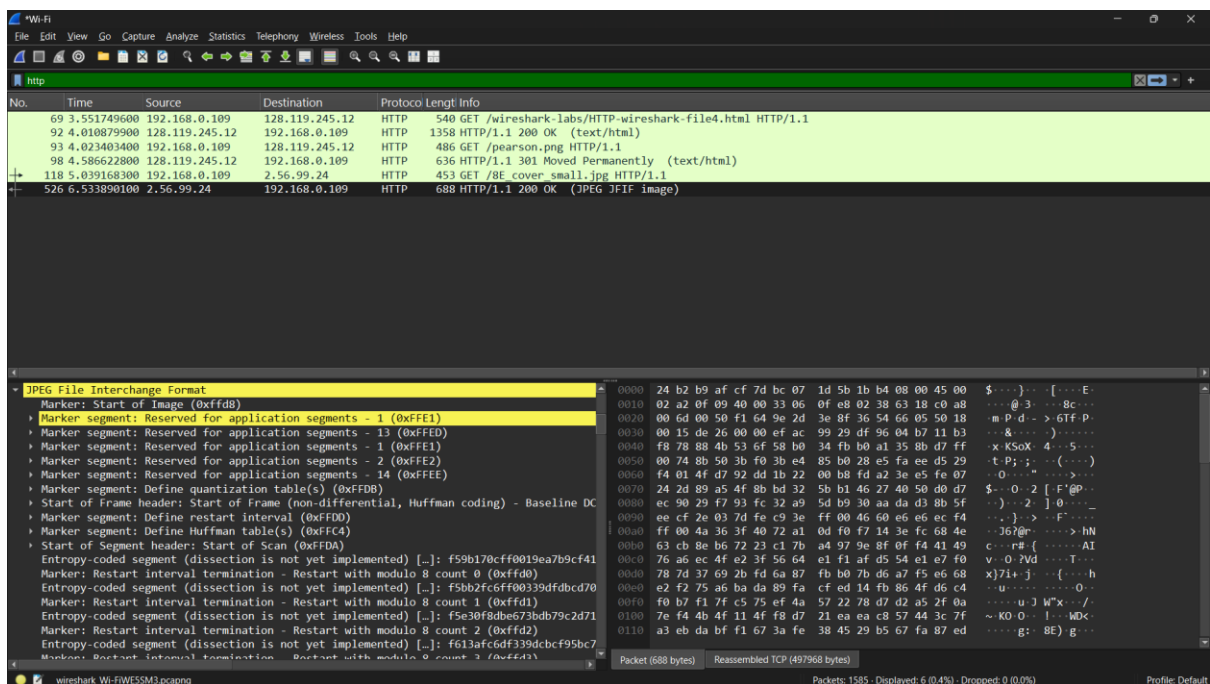
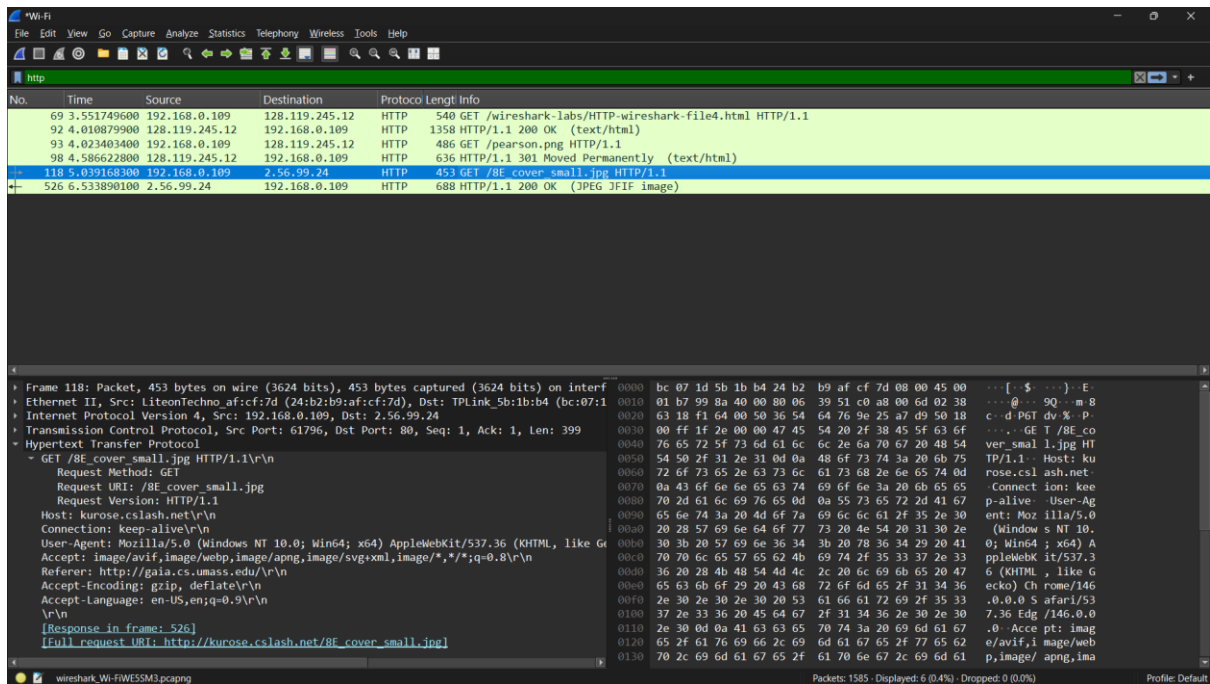
http

No.	Time	Source	Destination	Protocol	Length	Info
69	3.551749600	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
92	4.010879900	128.119.245.12	192.168.0.109	HTTP	1358	HTTP/1.1 200 OK (text/html)
93	4.023403400	192.168.0.109	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
98	4.586622800	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
118	5.039168300	192.168.0.109	2.56.99.24	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
526	6.533890100	2.56.99.24	192.168.0.109	HTTP	688	HTTP/1.1 200 OK (JPEG JFIF image)

Frame 118: Packet, 453 bytes on wire (3624 bits), 453 bytes captured (3624 bits) on interface 0
Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:11:5b:1b:b4)
Internet Protocol Version 4, Src: 192.168.0.109, Dst: 2.56.99.24
Transmission Control Protocol, Src Port: 61796, Dst Port: 80, Seq: 1, Ack: 1, Len: 399
Hypertext Transfer Protocol
GET /8E_cover_small.jpg HTTP/1.1
Request Method: GET
Request URI: /8E_cover_small.jpg
Request Version: HTTP/1.1
Host: kurose.cslash.net
Connection: keep-alive
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/69.0.3497.100 Safari/537.36
Accept: image/avif,image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8
Referer: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9
[Response in frame: 526]
[Full request URI: http://kurose.cslash.net/8E_cover_small.jpg]

Wireshark Wi-FiWESSM3.pcapng

Packets: 1585 - Displayed: 6 (0.4%) - Dropped: 0 (0.0%)



HTTP Authentication

Lab2-4 file: Embedded URLs x gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.html

https://gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.html

This page is password protected! If you're seeing this, you've downloaded the page correctly
Congratulations!

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
2854	102.866788	192.168.0.109	34.104.35.123	HTTP	314	HEAD /edgedl/diffgen-puffin/lme1glejhemejginboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9efe4ecca7919b5df7924...
2921	102.916006	34.104.35.123	192.168.0.109	HTTP	675	HTTP/1.1 200 OK
2950	102.943442	192.168.0.109	34.104.35.123	HTTP	365	GET /edgedl/diffgen-puffin/lme1glejhemejginboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9efe4ecca7919b5df7924...
3038	103.014620	34.104.35.123	192.168.0.109	HTTP	845	HTTP/1.1 200 OK

Frame 2854: Packet, 314 bytes on wire (2512 bits), 314 bytes captured (2512 bits) on interface 0

Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink 5b:1b:b4 (bc:07:1d:5b:1b:b4)

Internet Protocol Version 4, Src: 192.168.0.109, Dst: 34.104.35.123

Transmission Control Protocol, Src Port: 57703, Dst Port: 80, Seq: 1, Ack: 1, Len: 260

Hypertext Transfer Protocol

HEAD /edgedl/diffgen-puffin/lme1glejhemejginboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9efe4ecca7919b5df7924...
Request Method: HEAD
Request URI: /edgedl/diffgen-puffin/lme1glejhemejginboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9efe4ecca7919b5df7924...
Request Version: HTTP/1.1
Connection: Keep-Alive
Accept: */*
Accept-Encoding: identity
User-Agent: Microsoft BITS/7.8
Host: edgedl.me.gvt1.com

[Response in frame: 2921]
[col] request URI: http://edgedl.me.gvt1.com/edgedl/diffgen-puffin/lme1glejhemejginboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9efe4ecca7919b5df7924...

wireshark-Wi-FiC43MM3.pcapng

Packets: 3602 - Displayed: 4 (0.1%)

Profile: Default

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
2864	102.866788	192.168.0.109	34.104.35.123	HTTP	314	HEAD /edgedl/diffgen-puffin/lme1glejhomejginpboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9ef4ecca7919b5df7924...
2921	102.916006	34.104.35.123	192.168.0.109	HTTP	675	HTTP/1.1 200 OK
2950	102.943442	192.168.0.109	34.104.35.123	HTTP	365	GET /edgedl/diffgen-puffin/lme1glejhomejginpboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9ef4ecca7919b5df7924...
3038	103.014620	34.104.35.123	192.168.0.109	HTTP	845	HTTP/1.1 200 OK

content-length: 42278\r\n
[Content length: 42278]
content-security-policy: default-src 'none'\r\n
server: Google-Edge-Cache\r\n
x-content-type-options: nosniff\r\n
x-frame-options: SAMEORIGIN\r\n
x-identity-content-length: 42278\r\n
x-xss-protection: 0\r\n
x-request-id: 9ad163f2-2516-427c-8eac-aadaf615c454\r\n
date: Thu, 02 Apr 2026 14:58:54 GMT\r\n
age: 47287\r\n
last-modified: Thu, 02 Apr 2026 14:55:43 GMT\r\n
etag: "5b006d1"\r\n
content-type: application/octet-stream\r\n
vary: Accept-Encoding\r\n
alt-svc: h3=":443"; ma=2592000, h3-29=":443"; ma=2592000\r\n
cache-control: no-transform,public,max-age=86400\r\n
\r\n
[Request in frame: 2864]
(Time since request: 40.319200 milliseconds)

0000 24 b2 b9 af cf 7d bc 07 1d 5b 1b b4 08 00 45 00 \$...-...-...-...E...
0010 02 95 87 72 00 00 3b 06 ee f8 22 68 23 7b c0 a8 ...r...: ...hB{...
0020 00 6d 00 50 e1 67 f3 32 78 bf 7c 74 a0 91 50 18 ...m P g 2 x | t...P...
0030 04 1a 16 87 00 00 48 54 54 50 2f 31 2e 31 20 32 ...-...-HT TP/1.1 2...
0040 30 30 20 4f 4b 0d 0a 61 63 63 65 70 74 2d 72 61 00 OK ...a ccept-ra...
0050 6e 67 65 73 3a 20 62 79 74 65 73 0d 0a 63 6f 6e nges: by tes -con...
0060 74 65 6e 74 2d 64 69 73 70 6f 73 69 74 69 6f 6e tent-dis position...
0070 3a 20 61 74 74 61 63 68 6d 65 6e 74 0d 0a 63 6f : attach mont...co...
0080 6e 74 65 6e 74 2d 6c 65 6e 67 74 68 3a 20 34 32 ntent-le ngth: 42...
0090 32 37 38 0d 0a 63 6f 6e 74 65 6e 74 2d 73 65 63 278...con tent-sec...
00a0 75 72 69 74 79 2d 70 6f 6c 69 63 79 3a 20 64 65 urity-po licy: de...
00b0 66 61 75 6c 74 2d 73 72 63 20 27 6e 6f 6e 65 27 fault-sr c 'none'
00c0 0d 0a 73 65 72 76 65 72 3a 20 47 6f 6f 6f 6c 65 ...server : Google...
00d0 2d 45 64 67 65 2d 43 61 63 68 65 0d 0a 78 2d 63 -Edge-Ca che -x-c...
00e0 6f 6e 74 65 6e 74 2d 74 79 70 65 2d 6f 70 74 69 ontent-t ype-opti...
00f0 6f 6e 73 3a 20 6e 6f 73 6e 69 66 66 0d 0a 78 2d ons: nos niff -x-...
0100 66 72 61 6d 65 2d 6f 70 74 69 6f 6e 73 3a 20 53 frame-op tions: S...
0110 41 4d 45 4f 52 49 47 49 4e 0d 0a 78 2d 69 64 65 AMEORIGI N -x-ide...
0120 6e 74 69 74 79 2d 63 6f 6e 74 65 6e 74 2d 6c 65 ntity-co ntent-le...
0130 6e 67 74 68 3a 20 34 32 32 37 38 0d 0a 78 2d 78 ngth: 42 278 -x-x...

Packets: 4776 · Displayed: 4 (0.1%) · Dropped: 0 (0.0%) Profile: Default